Viscosity For Dispersed Graphene Oxide In Coconut Oil

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***Abstract*— The use of Graphene Oxide as a support material for the dispersion of nano materials in base fluid to prepare a nano fluids. Nano fluids, the fluid suspensions of nano materials, have shown many interesting properties, and the distinctive features offer unprecedented potential for many applications. This paper summarizes the recent progress on the study of nano fluids, such as the preparation methods, the evaluation methods for the stability of nano fluids. In the present study, we describe the synthesis procedure of Graphene Oxide (GO) and GO nano fluid preparation. The prepared nano fluids with various concentration of GO (0-0.4 g) was studied by ultrasonics studies such as velocity, acoustical impedance, adiabatic compressibility and inter molecular free length. The velocity results that at lower concentration, the pseudo plastic property of the nano fluid is due to non-Newtonian behavior of the base fluid, but at higher concentrations, nano particle–fluid interaction.**

Keywords—Graphene Oxide;Nano fluid; Velocity; Adiabatic Compressibility;Micro Raman

I. INTRODUCTION

A suspension of nano-size particles in conventional fluids, such as water, ethylene glycol, and oil, is called nano fluid. Nano fluids can be considered as an innovative group of engineering resources in a formula of suspensions of nano particles in its base liquid. Nano fluids thermophysical and rheological properties are affected from various particles', its concentrations, sizes, and base fluids etc. For that reason, the best way for an accurate solution is to measure these properties directly due to the possible inconsistency of the empirical correlations in the literature [1]. Graphite oxide, formerly called graphitic oxide or graphitic acid is a compound of carbon, oxygen, and hydrogen in variable ratios, obtained by treating graphite with strong oxidizers. Nonlinear optical materials are of great importance for ultrafast photonics and optoelectronics. Recently, the giant optical nonlinearities of graphene oxide (GO) has been proved useful for a number of applications. One of the most intriguing and unique properties of GO is that its electrical and optical properties can be tuned dynamically by manipulating the content of oxygen containing groups through either chemical or physical reduction methods .

II. EXPERIMENTAL PROCEDURE

Preparation Of GO and Nano Fluid:

Graphene oxide was prepared by the modified Hummers method. In brief 2g of natural flake graphite was added to 46ml of concentrated H_2SO_4 in a 250ml round bottom flask maintained at 0-5^{0}C and stirred at this temperature to produce a homogeneous suspension nearly 6g of $KMnO_4$ was slowly added to this suspension and further stirred for 2hrs.The flask was then transferred to a pre heated oil bath maintained at 35^{0}C and stirred for 12hrs. Then 92ml of DI water was carefully added to 250ml of DI water followed by addition of 30% H_2O_2 until the color changed to bright yellow. 5%HCl solution was added and washed several times with DI water to remove excess acid. The removal of HCl and H_2SO_4 was confirmed by silver nitrate and barium chloride solution. The GO dispersion in water and then sonicated the GO solution and dried the GO.

There are two kinds of preparation method for nano fluids: One step and the two-step methods [2]. The two-step method forms nano particles using physical or chemical synthesis techniques and disperses them in the base fluids. A promising one step method, physical synthesis technique, is the Electrical Explosion of Wire in liquid. In this work, GO/ Coconut oil mixtures have been experienced for the nano particle volume fractions (0,0.1,0.2,0.3,0.4 g). GO nano particles' size distribution was between 10 and 30 nm. To make the nano fluid at the necessary concentration, volume fractions were decided primarily.

III. RESULT AND DISCUSSION

A. FT-IR ANALYSIS

FTIR spectrum was recorded for the nano particle using Shimadzu 4000 spectrometer instrument in the range of 4000-400 cm^{-1} at a resolution of 2 cm^{-1}. FTIR spectroscopy is an analytical technique

used to identify organic polymers and nano particles. The characteristic absorption peaks of GO is shown in fig.1. The presence of strong vibrational peaks corresponding to oxygen containing groups at 1729cm^{-1}. The vibrational peaks at 1620cm^{-1} is corresponding to aromatic C=C stretching vibration [3]. The vibrational peak 1398cm^{-1}, 1227cm^{-1}, 872cm^{-1} is denotes that O-H bending, C-O-C of an epoxy group and alkoxy group. The broader peak around 3405cm^{-1} is assigned to hydroxyl group. These signals indicated that the rich oxygen-based functional groups were successful grafted on surface.

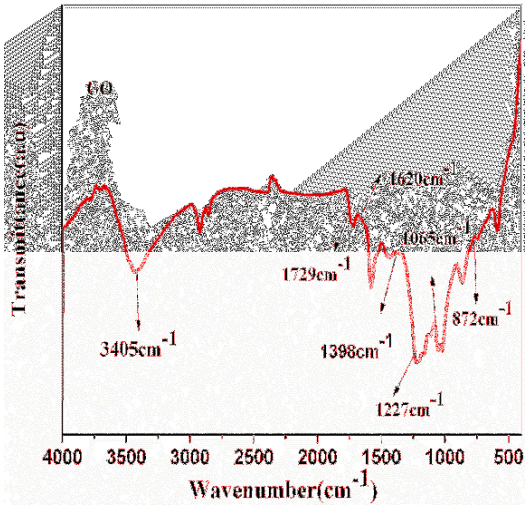

Fig.1 FTIR spectra of Graphene Oxide

B. Micro Raman Analysis

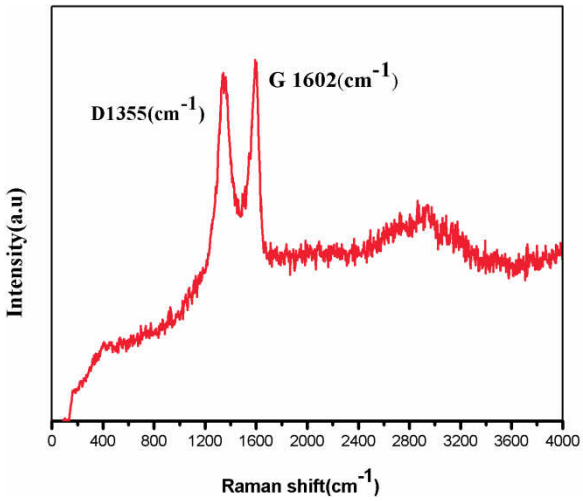

Fig.2 Micro Raman spectra of Graphene Oxide

Raman spectroscopy is an excellent tool to determine the carbon based materials. As a pseudo-ordered structure, the surface of GO is covered non uniformly with those oxygenated groups leaving approximately 2-3 nm. SP^2 carbon clusters isolated within the SP^3 carbon matrix, as can be easily verified by Raman spectroscopy [4]. In fig.2 the Raman spectrum of GO basically consist of a broad D peak 1355cm^{-1} and G peak 1602cm^{-1} arises from SP^2 bonded ordered graphite carbon. The ratio of ID/IG value is 1.36, indicating large amount of defects within the crystal lattice.

C. Ultrasonic velocity

Ultrasonic technique allows us to investigate the modifications induced by the nano particle in the local structure of the solvent, that in turn to generate changes in the adiabatic compressibility of system [5]. It is calculated from the speed of sound and the density of the medium using Newton and Laplace.

$$\beta = 1/v^2 \qquad (1)$$

The ultrasonic velocity of GO nano fluid increases as function of concentration at room temperature, the values are shown in table.1. But at lower concentrations 0.2wt% there is a sudden decrease in velocity followed by slight increment with loading of nano particle as shown in fig.3. With the addition of nano particle to the base fluid the particle fluid interaction increases which in turn reduces ultrasonic velocity. Here the concentration of 0.2wt% can be considered as critical concentration above which particle-particle interaction dominates the particle-fluid concentration. The nano particle loading there is a possibility for decreasing rate of occurrence of Brownian motion of fluid molecule, along with the formation for resistive surface layer that can persuade a decrease in velocity.

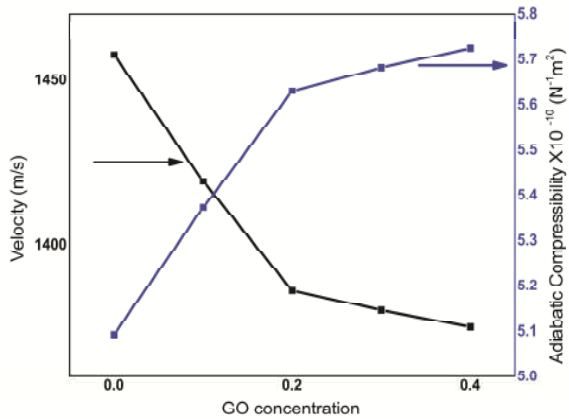

Fig.3 Velocity and Adiabatic compressibility Studies of nano fluid (GO/Coconut Oil)

D. Adiabatic Compressibility

The adiabatic compressibility is the sensitive to the molecular and the modification local structure of due to the addition of solute. The variation of adiabatic compressibility with concentration is illustrated in fig 3. The change in the compressibility values of the nano fluids with respect to that of the base fluids is

found to be negligible at low concentration. Adiabatic compressibility can be expressed by

$$\beta = 1/U^2\rho \quad (2)$$

Table.1 Velocity and Adiabatic Compressibility of Nano fluid for GO various concentration.

S.No	Concentration of Nano Fluid		Velocity (ms^{-1})	Adiabatic compressibility $x10^{-10}(N^{-1}m^2)$
	Coconut Oil(ml)	*GO(g)*		
1	10	0	1457.6	5.093
2	10	0.1	1419.2	5.373
3	10	0.2	1386	5.630
4	10	0.3	1380	5.682
5	10	0.4	1375	5.724

E. Acoustical impedance

The acoustical impedance is the impedance offered to the sound wave by the components of the mixture of solution from the observation the result it has been explained that the impedance system decreases with increase of nano particle colloidal suspension. The characteristic acoustic impedance was calculated for all samples using the relation.

$$Z = \rho v \quad (3)$$

The variation of acoustic impedance with concentration is illustrated in fig4. There is a small increase in acoustic impedance at lower concentration but as concentration increases beyond 0.2wt%. The acoustic impedance is the coefficient of proportionality between pressure and velocity of the particles in the sound wave. This explains a significant reduction in particle velocity beyond critical concentration.

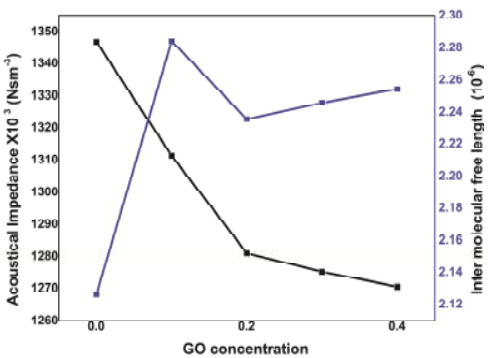

Fig.4 acoustical impedance & inter molecular free length Of GO/ Coconut oil nano fluid

F. Inter molecular free length of nano fluid

$$L_F = KT\beta \quad (4)$$

Here K ((93.875+0.345) $X10^{-8}$) is a Jacobin constant.

From table.2 the positive excess value of free length should be attributed to the dispersive forces and negative excess values should be due to charge transfer and hydrogen bond formation. In the present study the positive contribution of free length in all the systems which prevails the existence of dispersive force acting in the nano fluid system.

S.No	Concentration of Nano Fluid		Acoustic Compressibility $X10^{-10}$ (N/m^{-2})	Inter Molecular free length ($X10^6$)
	Coconut Oil(ml)	*GO(g)*		
1	10	0	1346.8	2.1263
2	10	0.1	1311.3	2.2839
3	10	0.2	1281.0	2.2355
4	10	0.3	1275.1	2.2458
5	10	0.4	1270.5	2.2541

Table.2 Nano fluids acoustic compressibility and inter molecular free length value for various concentration of GO.

IV. CONCLUSION

Graphene oxides were synthesized via modified Hummers method. Nano fluids of various concentrations of GO in coconut oil as a base fluid have been prepared and they have been studied by various characterization successfully. The investigation leads to the understanding of the molecular interactions taking place between the nano particles and the fluid. It is found that the GO nano particles interact well with the coconut oil. Beyond 0.2 wt% the particle-particle interaction becomes dominant than the particle fluid interaction and the positive contribution of free length in all prepared nano fluid system. From the results that at lower concentration the pseudo plastic property of the nano fluid is due to non-Newtonian behavior of the base fluid, but at higher concentrations, nano particle–fluid interaction.

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